

SAIKHAN-OVOO, Mongolia

WMO: 443360

Lat: 45.4611N

Lon: 103.9000E

Elev: 4318

StdP: 12.54

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-21.2	-32.2	1.0	-22.9	-28.1	1.3	-19.0	26.6	11.9	22.3	13.3	3.0	310	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	90.9	61.2	87.7	60.1	84.6	58.6	64.9	83.0	62.9	80.5	61.3	77.5	8.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.6	89.4	68.5	57.4	82.5	67.8	55.2	76.2	68.1	32.4	82.5	30.7	81.0	29.4	77.3	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.7	23.1	20.4	DB	-29.6	95.8	5.7	3.0	-33.7	98.0	-37.1	99.7	-40.3	101.4	-44.5	103.6	(4)
(5)			WB	-29.7	67.8	5.6	2.8	-33.7	69.8	-37.0	71.5	-40.2	73.1	-44.3	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	-1.3	6.9	25.6	42.9	55.3	66.5	72.0	68.1	55.8	39.0	19.2	3.5
	DBStd	26.93	9.54	11.34	10.59	9.19	8.75	6.92	6.05	6.47	8.16	8.70	11.21	9.13
	HDD50	6603	1590	1207	757	245	48	0	0	0	38	350	925	1443
	HDD65	10375	2055	1627	1221	662	317	63	11	41	289	806	1375	1908
	CDD50	2206	0	0	1	33	212	496	681	562	212	9	0	0
	CDD65	504	0	0	0	0	17	109	227	138	13	0	0	0
	CDH74	5594	0	0	0	8	316	1323	2372	1388	187	0	0	0
	CDH80	1977	0	0	0	1	75	464	948	463	26	0	0	0
Wind	WSAvg	7.0	4.2	5.0	8.1	9.3	10.2	8.7	7.7	7.0	7.0	6.9	5.5	4.1
Precipitation	PrecAvg	5.2	0.0	0.0	0.1	0.2	0.4	1.1	1.5	1.1	0.4	0.2	0.1	0.1
	PrecMax	8.9	0.2	0.2	0.5	0.5	2.0	3.8	3.8	4.6	1.3	0.5	0.4	1.1
	PrecMin	2.5	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.1	0.0	0.1	0.2	0.4	0.9	1.0	0.9	0.4	0.2	0.1	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.1	41.6	61.0	74.5	86.3	93.4	97.2	92.4	83.7	69.3	52.1	33.2
		MCWB	22.8	30.6	40.9	50.6	55.7	60.8	64.8	62.1	56.2	47.6	38.0	25.6
	2%	DB	23.4	35.3	54.7	69.8	81.7	89.0	92.1	88.6	79.1	63.9	45.9	27.1
		MCWB	18.2	26.6	37.8	47.0	53.5	59.3	62.6	60.5	54.6	45.4	34.1	21.0
	5%	DB	19.0	30.4	49.4	65.1	77.1	85.3	88.5	85.2	75.3	59.5	41.1	22.8
		MCWB	15.0	23.2	34.9	44.8	51.3	57.6	61.3	59.2	53.1	42.7	31.2	18.1
	10%	DB	14.6	25.8	44.5	60.6	73.4	81.6	85.4	81.7	71.4	55.3	36.4	18.5
		MCWB	11.4	20.3	32.2	42.4	49.7	56.2	60.0	57.7	51.5	40.7	28.3	14.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.2	30.7	41.8	52.6	57.8	63.9	68.4	65.9	59.3	49.6	38.3	25.6
		MCDB	29.7	40.1	58.9	72.0	82.1	86.4	86.8	83.9	76.7	66.7	51.4	32.6
	2%	WB	18.5	26.8	38.2	48.1	54.9	61.6	66.2	63.6	56.8	46.0	34.3	21.4
		MCDB	23.4	34.9	53.7	66.5	78.0	80.4	85.3	80.6	73.1	62.4	45.3	26.8
	5%	WB	15.0	23.6	35.2	45.3	52.5	60.2	64.2	61.9	54.8	43.3	31.3	18.2
		MCDB	19.0	30.0	48.8	63.9	74.6	77.8	81.4	77.5	71.3	58.1	40.6	22.7
	10%	WB	11.5	20.4	32.3	42.8	50.3	58.5	62.4	60.3	52.7	40.8	28.5	14.8
		MCDB	14.7	25.7	44.1	59.8	71.3	76.3	79.7	74.8	68.8	54.5	36.3	18.4
Mean Daily Temperature Range	5% DB	MDBR	24.4	28.1	27.6	25.4	24.8	23.0	21.0	21.4	23.8	25.8	24.3	22.8
		MCDBR	27.6	32.6	32.8	31.2	29.9	28.1	25.0	25.8	28.4	31.0	28.2	26.1
		MCWBR	23.3	25.3	20.2	17.1	14.4	11.6	8.9	9.6	12.5	17.4	19.1	21.3
	5% WB	MCDBR	27.6	31.4	31.9	29.3	27.8	23.6	21.1	21.6	25.0	29.4	27.5	25.6
		MCWBR	23.3	24.7	20.2	17.1	15.0	10.8	9.7	8.8	12.1	17.2	18.9	21.0
Clear-Sky Solar Irradiance	taub		0.250	0.277	0.319	0.362	0.380	0.398	0.385	0.364	0.327	0.298	0.265	0.249
	taud		2.356	2.294	2.213	2.147	2.150	2.190	2.285	2.323	2.372	2.384	2.386	2.353
	Ebn at Noon		278	288	288	283	281	276	278	280	283	277	269	265
	Edh at Noon		23	30	38	45	47	45	41	37	33	28	23	21
All-Sky Solar Radiation	RadAvg		641	999	1494	1910	2217	2210	2109	1916	1616	1156	734	543
	RadStd		56	86	56	79	107	109	84	92	51	50	61	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	-4.27	N/A	-1.62	-1.89	N/A	N/A	N/A	N/A

Nomenclature: See separate page